

Transformer-Based Approaches to Rule Violation Detection in Online Communities

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The Problem: Volunteer Community Moderation



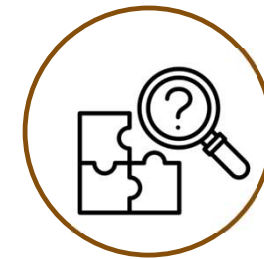
Community-specific rules

*Same comment means different things
in different communities*



Manual review at scale

26.6K communities
~400K comments posted every month
Moderation largely relies on volunteer administrators



Beyond toxicity

Violation depends on the specific rule

Estimated from Lemmy Statistics snapshots, May–July 2026

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Why community-specific rules matter?

User comment:

"Lighten up, psycho"

memes@lemmy.ml — REMOVED

Community Rule:

"Be civil and nice."

Calling someone a "psycho" is deemed uncivil.

RULE VIOLATION DETECTED

linux@lemmy.world — ALLOWED

Community Rule:

"No off-topic posts or spam."

Comment is on-topic and opinionated, so does not violate this rule.

NO VIOLATION



Prior work

- Much prior automated moderation work classifies predefined categories of harmful content, such as toxicity, insults, or threats.
- These predictions are typically based on the comment text alone, without conditioning on the community's own rule descriptions.
- More recent rule-aware approaches have expanded beyond toxicity, but open-ended selection from a dynamically provided rule set remains underexplored.

Examples: Wulczyn et al. (2017) on personal attacks; Park et al. (2021), NormVio; He et al. (2024), CPL-NoViD; Xin et al. (2024), rule-aware moderation.



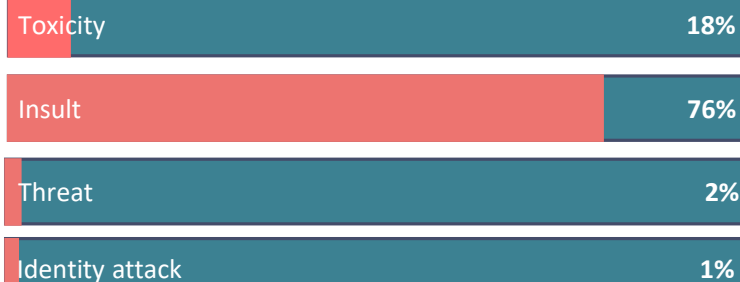
The Gap: Toxicity Scores vs. Rule Reasoning

Detoxify¹ (fixed toxicity categories)

Comment:

"Lighten up, psycho"

Scores over predefined outputs:



INSULT DETECTED

Rule-Aware Approach

Comment + Community Rules:

"Lighten up, psycho" +
Community rules: [Be civil | No repost]
No-violation option: [SAFE]

Model output (probability per rule):

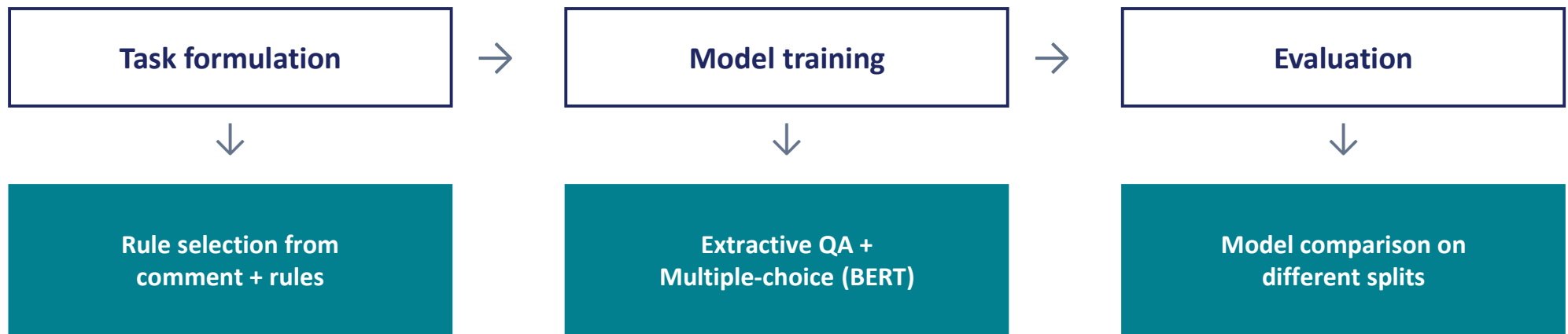


VIOLATED: "Be civil and nice."

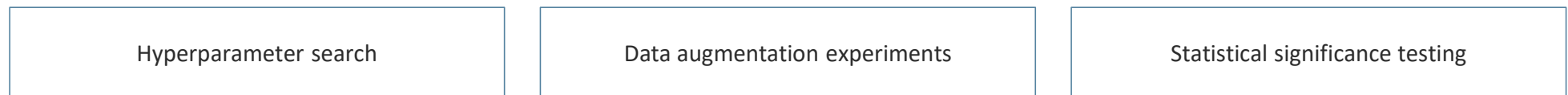
¹ Detoxify is a widely used open-source toxicity-classification library, selected as a recognizable example of comment-only moderation: approximately 113K PyPI downloads per month and 1.3K GitHub stars as of July 2026..



General Outline



ALSO CONDUCTED



The Raw Dataset

FIELDS USED BY THE MODEL

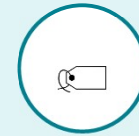
Field	Example value
content	"Lighten up, psycho."
community.name	memes
community.rules	{"0": "Safe", "1": "Be civil", "2": "No politics"}
applied_rule_text	"Be civil"

DATASET PROPERTIES



Federated platform

*Public moderation logs
Diverse communities with independently
defined rule sets*



Observed moderation action

*Recorded removal + applied rule provide
the label*



Rule sets vary

*2 to 11 rules
per community*



Special SAFE candidate

*Represents cases when no listed
community rule was violated*



Input Representation

Extractive QA

All rules + comment, one block of text:

Rules: **Be civil** No politics Safe
Content: "**Lighten up psycho.**"

↓ model finds where the rule begins & ends

"Be civil" found in the text

Multiple-Choice

Comment is compared to each rule separately:

"Lighten up psycho."

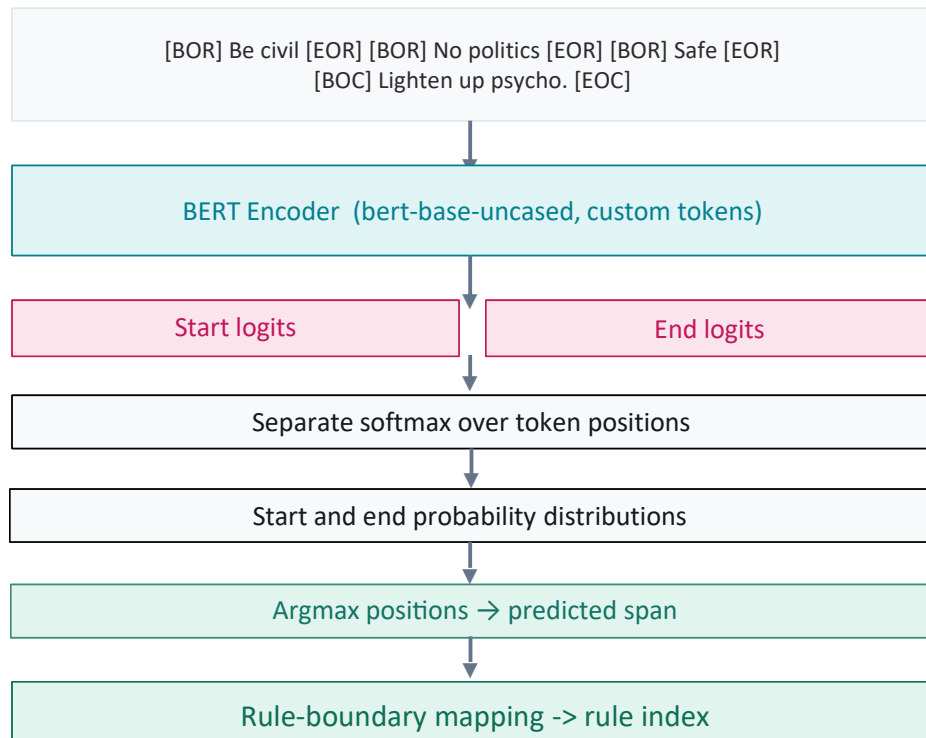
A	Be civil	81%
B	No politics	11%
C	Safe	8%

"Be civil" — highest score wins



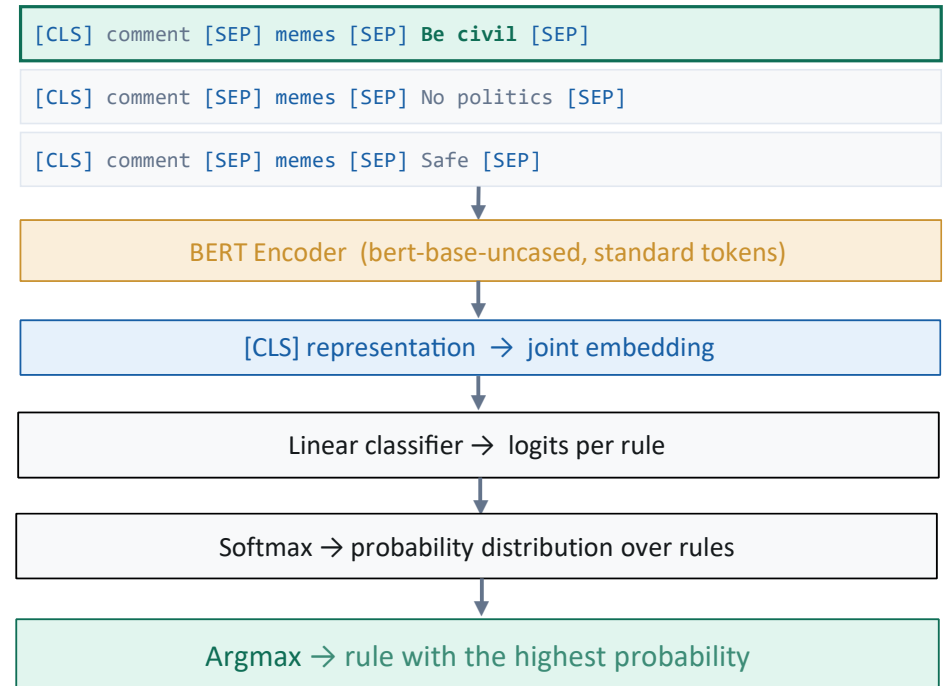
Model Architecture (BERT-base-uncased)

EXTRACTIVE QA



MULTIPLE-CHOICE

One sequence per candidate rule:



Results

STRATIFIED (Macro-F1)		UNSEEN RULES (Macro-F1)		UNSEEN COMMUNITIES (Macro-F1)	
Multiple-choice	0.691	Multiple-choice	0.193	Multiple-choice	0.328
Extractive QA	0.588	Extractive QA	0.144	Extractive QA	0.324
Linear SVM	0.173	Linear SVM	0.196	Linear SVM	0.116
Random	0.083	Random	0.076	Random	0.055

McNEMAR'S TEST — MC vs EQA

Stratified MC only: 296 EQA only: 245 $\chi^2 = 4.621, p = 0.032$	Unseen rules MC only: 15 EQA only: 2 $\chi^2 = 8.471, p = 0.004$	Unseen communities MC only: 106 EQA only: 208 $\chi^2 = 32.487, p < 0.001$
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Findings



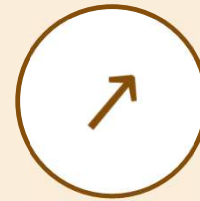
**Rule-aware modeling
helps in-distribution**

*Large advantage over comment-
only baselines on familiar rules*



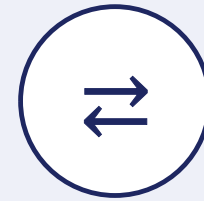
**Rule semantics do not
generalize**

*Linear SVM beats
transformers on unseen rules*



**Cross-sommunity transfer
is more tractable**

*New communities are easier
than entirely new rule
descriptions*



**Aggregate scores hide
error structure**

*Errors should be analyzed by
rule frequency, macro-
category, and default
prediction*



Conclusion



Rule-aware modeling helps in familiar settings

On the stratified and unseen-community splits, transformer models outperform comment-only baselines, showing that community rule text provides useful information.



Zero-shot transfer to new rules remains unresolved

On unseen rule descriptions, Linear SVM performs comparably to multiple-choice and outperforms extractive QA. This suggests possible reliance on familiar wording, but moderator and community context may also explain the failure.



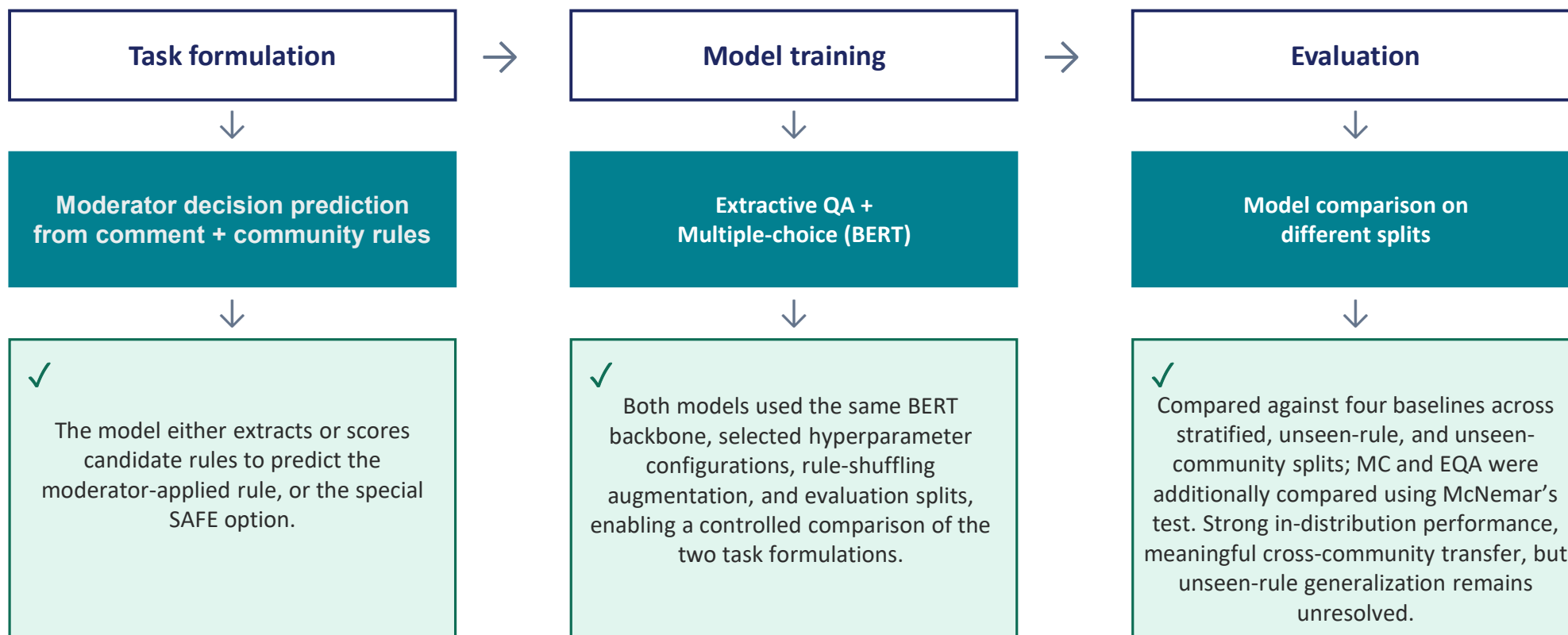
The next question is why models fail

Error analysis by rule frequency, semantic macro-category, and default prediction behavior is needed to understand failure on unseen rules.



Summary

Back to the outline of what we achieved at each stage



Data Augmentation Results

TESTED

01	Rule shuffling Randomly permutes rule order at each training step
02	Rule exclusion Removes non-violated rules from candidate list
03	Number permutation Perturbs numerical expressions in comments
+	Combined strategies All three strategies applied simultaneously

FOUND

✓	Rule shuffling Consistent +0.01–0.015 macro-F1 on unseen rules across both configurations
✗	Rule exclusion Inconsistent — helps with $lr=1e-5$ (+0.03), hurts with $lr=2e-5$ (−0.049)
✗	Number permutation Weakest results overall — adds noise without meaningful improvement
✗	Combined strategies No improvement over shuffling alone — macro-F1 dropped 0.676 → 0.526



Future Work

→ Zero-shot and few-shot rule generalization

Explore contrastive objectives and instruction-tuned models to improve generalization to previously unseen rule descriptions.

→ Stronger pretrained models

Evaluate RoBERTa, DeBERTa, and instruction-tuned models that may better capture the semantics of new rule descriptions.

→ Ensemble approach

Combine extractive and multiple-choice predictions: on unseen communities, EQA correctly classified 208 examples missed by MC, while MC uniquely classified 106.

→ Deployment and real-word integration

Explore a lightweight, locally deployable moderation assistant that each community can adapt to its own rules with minimal labeled data.

